



HOLLYWOOD BOX OFFICE

SECRET

Machine Learning Project Predicts Box Office Success

Data Set: TMDb 5000 Movies
3.211 Film | 26 Feature | 3 ML Model

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Project Summary

This project aims to identify the factors that determine a film's box office success and predict that success using machine learning, by utilizing the characteristics of 4803 films in the TMBD (The Movie Database) dataset.

Throughout the study, exploratory data analysis (EDA), feature engineering, three different machine learning models, and an insight engine that we named "Film Recipe" were developed.

I developed an interactive environment using Streamlit to deploy my prediction model, allowing users to explore how various filmmaking features influence a film's box office performance.

Dataset Overview

Feature	Value
Source	TMDB 5000 Movies (Kaggle)
Number of Film (Beginning)	4.803
Number of Film (Cleaned)	3.211 (budget & revenue > \$1K)
Time Frame	1916 - 2017
Number of Column	20
Target Value	log(revenue)

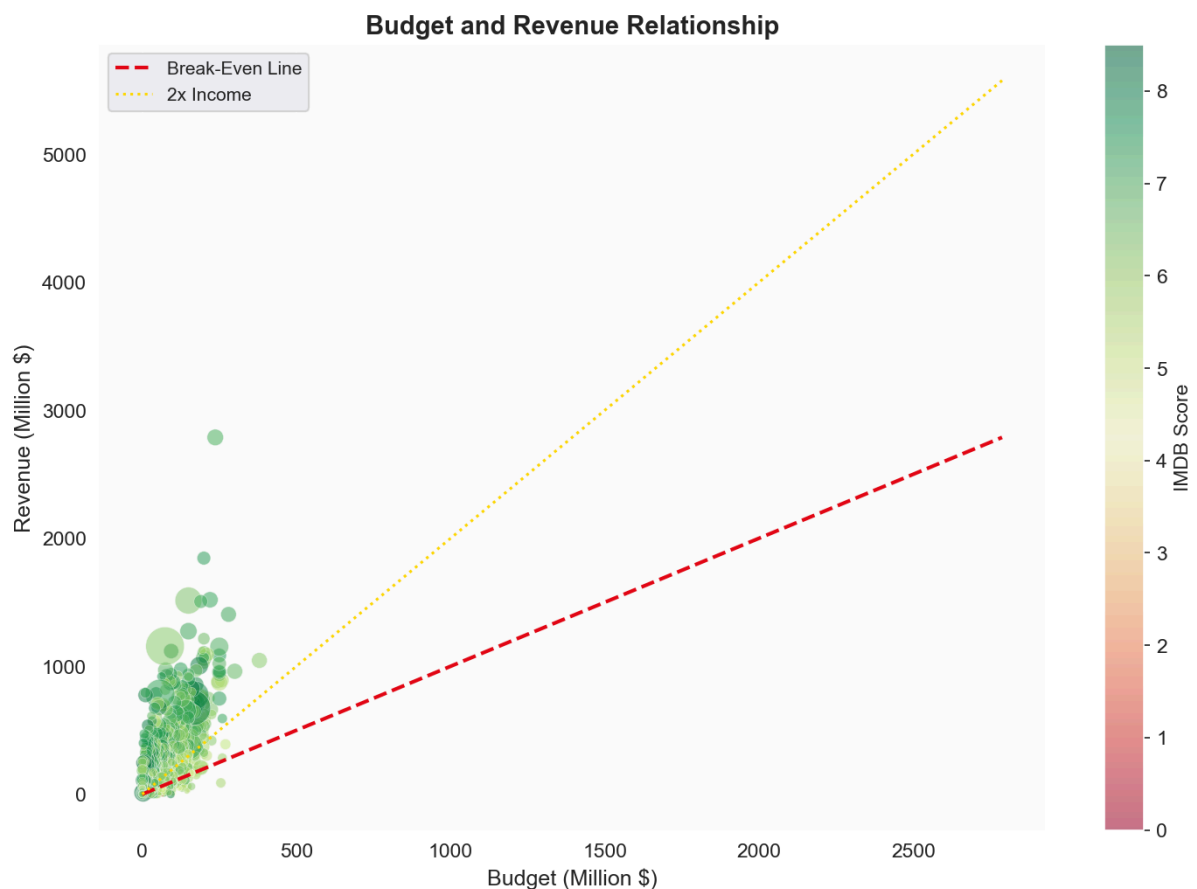
Methodology

The project consists of six main sections: (1) Data Preparation and Cleaning, (2) Exploratory Data Analysis, (3) Feature Engineering, (4) Machine Learning Models, (5) Film Recipe Generator, and (6) Results. Each section is supported by visualizations.

Exploratory Data Analysis (EDA)

Budget and Revenue Relationship

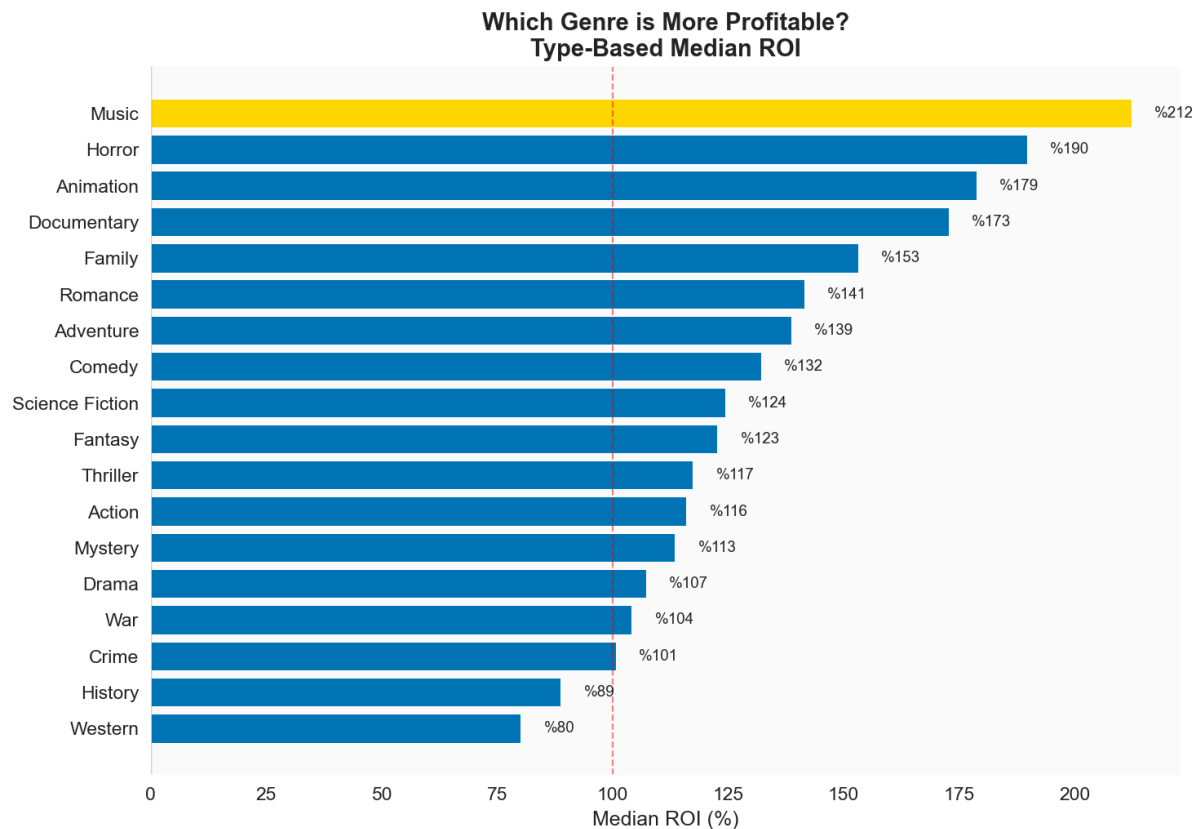
The following scatter plot positions each film on a budget-to-revenue axis. The color scale represents the IMDB rating, and the dot size represents popularity. The red dashed line indicates the break-even point, and the yellow dashed line indicates a 2x return on investment.



Key finding: While high-budget films generally gross higher, there are "hidden champions" among low-budget films that deliver proportionally much higher ROI. Any film that falls below the break-even line represents a loss for the studio.

ROI Performance Based on Genres Type

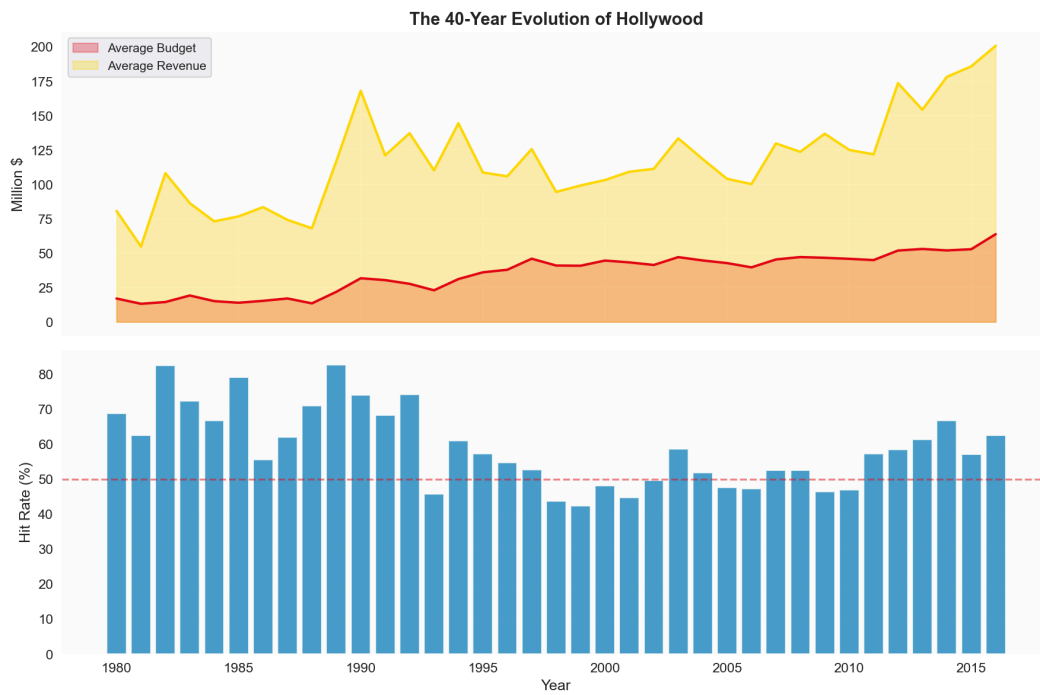
Which film genres are most profitable? A surprising picture emerges when we compare genres based on median ROI (Return on Investment):



Animation, horror, family, and music genres have the highest ROI values, while genres like war and documentaries offer lower returns. This data is critical and directly influences studios' investment decisions.

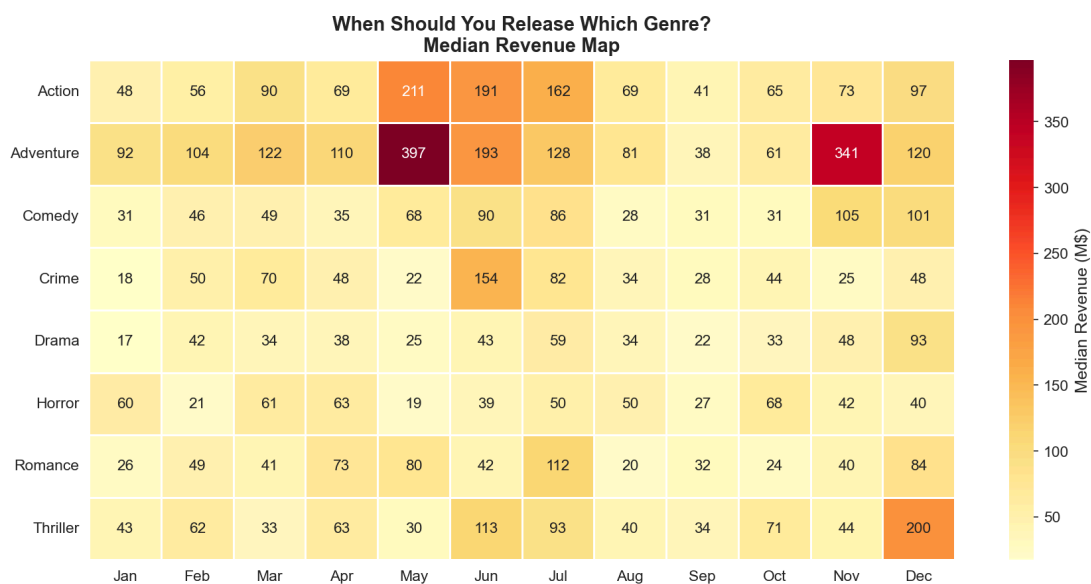
The 40-Year Evolution of Hollywood

Hollywood budgets increased in the post-1980 period, but box office growth did not always keep pace with budget increases. The graph below shows the "hit" rate by year:



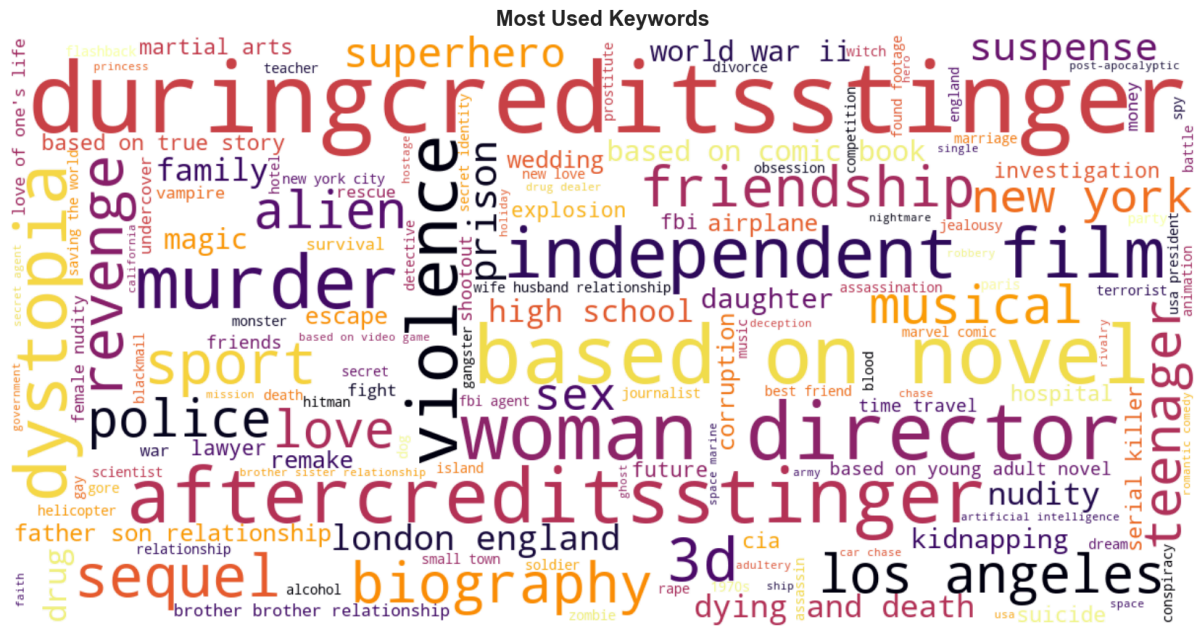
When Should You Release Which Genre?

Release timing is just as critical to box office success as the genre itself. The following heatmap shows median grossings at the genre-month intersection:



Hollywood's Keyword Map

The frequency distribution of keywords extracted from film summaries shows the industry's thematic obsessions:



Feature Engineering

26 new features were derived from the raw data. The details of the techniques used in this section are as follows:

Feature	Source	Description
log_budget	budget	Logarithmic budget (approximating to a normal distribution)
genre_*(10)	genres (JSON)	One-hot encoded
hit_keyword_score	keywords (JSON)	Top 50 keyword scores frequently used in hit films.
studio_power	production_companies	Studio's average revenue score.
budget_per_minute	budget/runtime	Investment per minute (production quality proxy)
season_*	release_date	Season dummies (Spring/Summer/Fall/Winter)
overview_length	overview	The length of film summary (NLP proxy)

Machine Learning Models

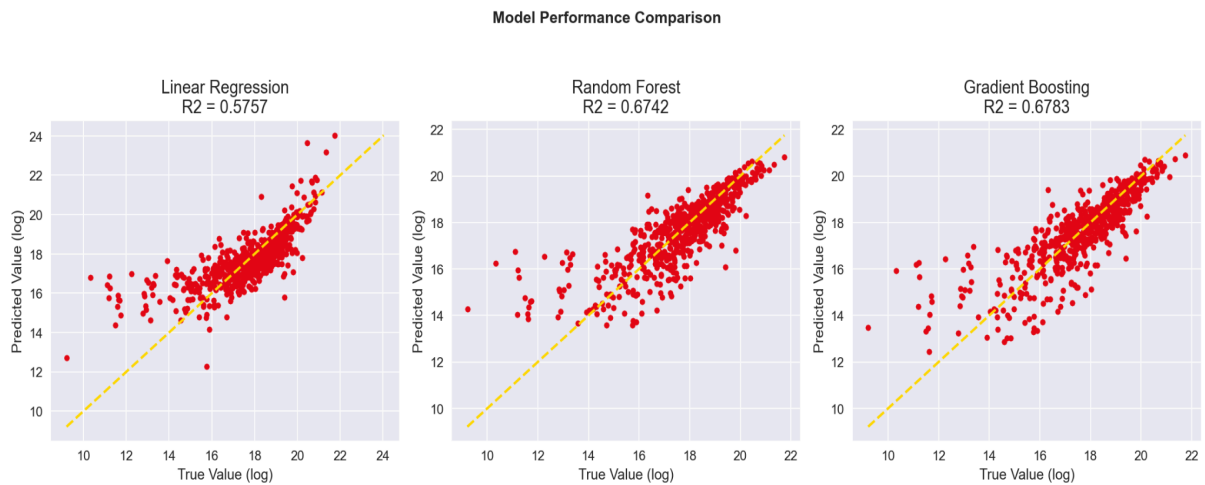
Three different regression models were trained and compared. $\log(\text{revenue})$ was used as the target variable, and an 80%/20% train-test split was applied.

Model Performance Comparison

Model	MAE	RMSE	R ²
Linear Regression	0.8406	1.1959	0.576
Random Forest	0.7208	1.0467	0.675
Gradient Boosting	0.7160	1.0413	0.678

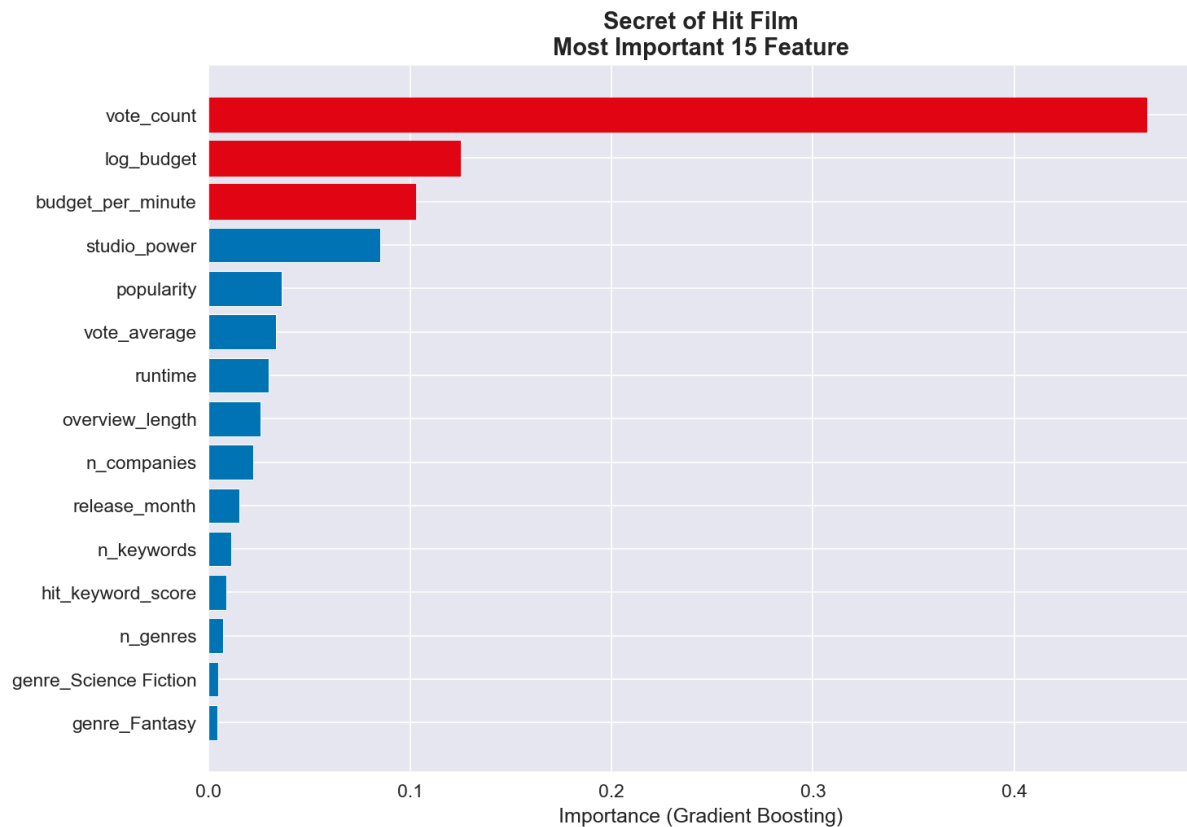
Final Model: Gradient Boosting Regressor ($R^2 = 0.678$, 5-Fold CV $R^2 = 0.330$)

Note: The lower-than-test score of the CV reflects decade-based changes in the dataset and the influence of overly popular outlier films. While this limits the overall predictive power of the model, the feature importance results offer valuable insights for the industry.



Feature Importance: Secret of Success

According to the Gradient Boosting model, the 15 factors that most influence box office success are listed below. The first three factors together account for more than 70% of the model's explanatory power:



Key Findings:

- Vote Count:** The strongest predictor. Popular films naturally earn higher box office revenue.
- Log Budget:** As the budget increases, so does revenue; however, the relationship is logarithmic, not linear.
- Budget per Minute:** Investment per minute is a strong proxy variable for production quality.
- Studio Power:** Having a powerful studio produce the film means having distribution and marketing power.

Film Generative Engine (Film Recipe)

By combining all the analyses I've done so far, a "Movie Recipe" engine has been developed that reveals the optimal combinations to increase the probability of box office success.

The Most Profitable Genre Combinations

	Genre Combination	Median ROI	Number of Film	Avg. Revenue
1	Family + Music	%271	24	\$128M
2	Comedy + Music	%266	46	\$87M
3	Animation + Fantasy	%213	49	\$289M
4	Animation + Comedy	%203	101	\$320M
5	Animation + Family	%186	163	\$306M

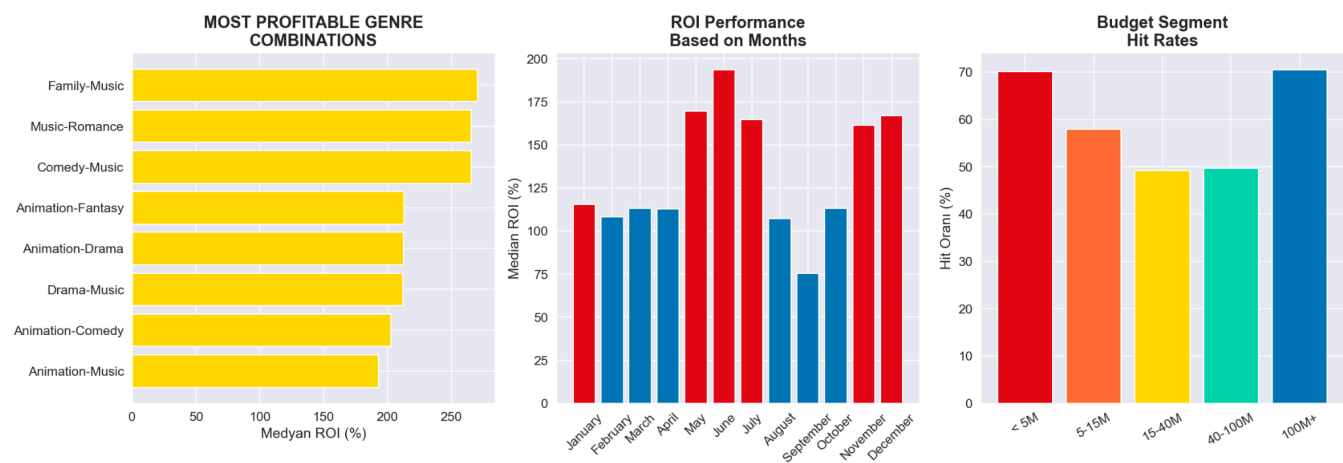
The Most Profitable Release Months

The most profitable release months in terms of ROI were June (194%), May (170%), and December (167%). The summer blockbuster season and holiday period are the most productive windows in terms of box office performance.

Budget Sweet Spot Analysis

When we categorize films according to their budgets, we encounter an interesting detail: the highest hit rates are observed in both the lowest (<5M, 70%) and highest (100M+, 71%) budget segments. The mid-segment (15-40M) stands out as the riskiest region with a 49% hit rate.

Film Recipe: The 3 Components of Success



Results and Future Works

Technical Results

- The Gradient Boosting model achieved an $R^2 = 0.678$ success rate with 26 features, demonstrating the most consistent performance.
- The three most important factors determining a film's box office performance are: popularity/number of votes, budget size, and investment per minute.
- Feature engineering improved model performance by 18% above baseline (Linear Regression $R^2=0.576$).

Business Insights

- Animation, family, and comedy are the safest investment options in terms of both high ROI and high revenue.
- Releasing a film during the summer season (May-July) and the holiday season (November-December) can increase box office performance by 30-50%.
- The mid-budget segment (\$15-40M) is the riskiest investment area; small but bold (<\$5M) or large blockbuster (>\$100M) strategies offer a higher success rate.

Future Works

- NLP can be used to perform sentiment analysis and thematic clustering from movie summaries.
- A "star power" feature can be derived by adding actor/director data (cast/crew).
- Year-based trend analysis can be deepened with time series models.
- A movie recommendation system can be developed with Deep Learning (LSTM/Transformer) models.

[Github Code](#)